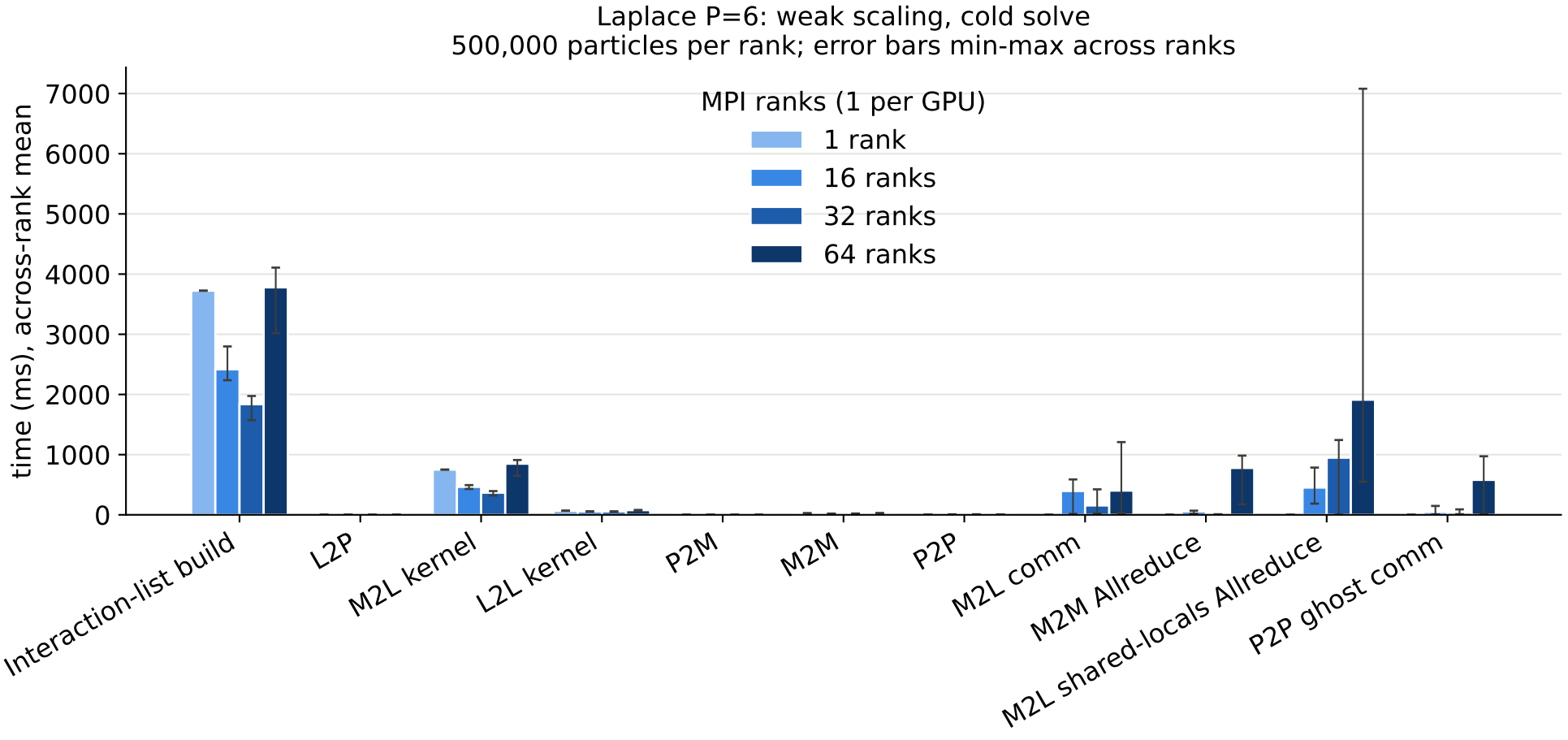
Laplace P=6: weak scaling, cold solve
500,000 particles per rank; error bars min-max across ranks
time (ms), across-rank mean
MPI ranks (1 per GPU)
1 rank
16 ranks
32 ranks
64 ranks
0
1000
2000
3000
4000
5000
6000
7000
Interaction-list build
L2P
M2L kernel
L2L kernel
P2M
M2M
P2P
M2L comm
M2M Allreduce
M2L shared-locals Allreduce
P2P ghost comm